Mth631

Q-1

Suppose $U_0 \in T$ is a limit point of T, G is continuous at U_0 and f is continuous at $X_0 = G(U_0)$. Then, show that $f \circ g$ is continuous at U_0? **Mark: 10**

Proof:-

Suppose $\varepsilon > 0$. Since f is continuous at $X_0 = G(U_0)$, there is an $\varepsilon_1 > 0$ such that

$$|f(X) - f(G(U_0))| < \varepsilon \longrightarrow (1)$$

if

$$|X - G(U_0)| < \varepsilon_1 \text{ and } X \in D_f \longrightarrow (2)$$

Since G is continuous at U_0, there is a $\delta > 0$ such that

$$|G(U) - G(U_0)| < \varepsilon_1$$ M.M

if $|U - U_0| < \delta$ and $U \in D_G$.

by taking $X = G(U)$ in (1) and (2) we get,

$$|h(U) - h(U_0)| = |f(G(U)) - f(G(U_0))| < \varepsilon$$

if

$$|U - U_0| < \delta \text{ and } U \in T$$

Note:-

"$\cup$" is a sign in Mathtype.

U_0

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Q.2

Show that $(0,0)$ is a critical point of the function $p(x,y) = x^2 - 2xy + y^2 - x^4 - y^4$ and it has positive semidefinite second differentials at $(0,0)$?

Soln:-

Given $p(x,y) = x^2 - 2xy + y^2 - x^4 - y^4$

$p_x(x,y) = 2x - 2y - 4x^3$

$p_y(x,y) = -2x + 2y - 4y^3$

M.m

* At point $(0,0)$

$p_x(0,0) = 0$ and $p_y(0,0) = 0$

So, $(0,0)$ is a critical point of p.

Now,

$p_{xx}(x,y) = 2 - 12x^2$

$p_{yy}(x,y) = 2 - 12y^2$

$p_{xy}(x,y) = -2$

$(d^2_{(0,0)} p)(x,y) = 2x^2 - 4xy + 2y^2$

$(d^2_{(0,0)} p)(x,y) = 2(x^2 - 2xy + y^2)$

$= 2(x-y)^2$

Hence it is +ve semidefinite.

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Q.3

Show that $(0,0)$ is a critical point of the function $p(x,y) = x^2 - 2xy + y^2 + x^4 + y^4$ and it has +ve semidefinite second differentials at $(0,0)$.

Soln:-

Given

$$p(x,y) = x^2 - 2xy + y^2 + x^4 + y^4$$

$$p_x(x,y) = 2x - 2y + 4x^3$$

$$p_y(x,y) = -2x + 2y + 4y^3$$

At point $(0,0)$

$$p_x(0,0) = 0 \quad \text{and} \quad p_y(0,0) = 0$$

So $(0,0)$ is a critical point of p.

Now

$$p_{xx}(x,y) = 2 + 12x^2$$

M.m

$$p_{yy}(x,y) = 2 + 12y^2$$

$$p_{xy}(x,y) = -2$$

$$(d^2_{(0,0)} p)(x,y) = 2x^2 - 4xy + 2y^2$$

$$(d^2_{(0,0)} p)(x,y) = 2(x^2 - 2xy + y^2)$$

$$(d^2_{(0,0)} p)(x,y) = 2(x-y)^2$$

Hence it is +ve semidefinite.

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Q.4

Define a locally integrable function. Also show that the function $f(x)=\log x$ is locally integrable on $[0,1)$.

Soln:-

Locally integrable function.

If f is a function, it is locally integrable on an interval I if f is integrable on every finite closed subinterval of I.

Proof:-

$$\int_1^0 \log x\,dx = x\ln x - x\Big|_1^0$$ M.M

$$= 0 - 1$$

$$= -1$$

Because it is unbounded as $x \to 0+$ since

$$\lim_{c\to 0+}\int_c^1 \log x\,dx = \lim_{c\to 0+}(x\log x - x)\Big|_c^1$$

$$= -1 - \lim_{c\to 0+}(c\log c - c) = -1$$

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Q-5

Suppose that $f(x,y) = \begin{cases} 0 & \text{if } x \text{ and } y \text{ are rational} \\ 1 & \text{if } x \text{ is rational and } y \text{ is irrational} \\ 2 & \text{if } x \text{ is irrational and } y \text{ is rational} \\ 3 & \text{if } x \text{ and } y \text{ are irrational} \end{cases}$

Soln:-

Let $P = \{R_1, R_2, \cdots R_K\}$

$M_j = \sup\{f(x,y) \mid X \in R_j\}$

$m_j = \inf\{f(x,y) \mid X \in R_j\}$ M.m

Let j be arbitrary in $\{1,2,\cdots K\}$. since R_j contains a point $(\hat{x}_j, \hat{y}_j)$ with $\hat{x}_j$ and $\hat{y}_j$ irrational $M_j = 3$. Hence, $\underline{\int_R} f(x,y)\,d(x,y) = 3(b-a)(d-c)$.

since R_j contains a point $(\tilde{x}_j, \tilde{y}_j)$ with $\tilde{x}_j$ and $\tilde{y}_j$ rational. $m_j = 0$ Hence,

$$\underline{\int_R} f(x,y)\,d(x,y) = 0$$

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Q.6

If a nonempty subset S of R^n is both open and closed, then $S = R^n$.

Soln:-

Let X_0 is in S. If $S \neq R^n$, there is an $X_1 \notin S$.

Let $H = \{\tau | (1-t)X_0 + tX_1 \in S \text{ for } 0 \leq 1 < \tau\}$

Since S is open it contains a neighborhood of X_0. So, $H \neq \phi$.

since X_1 is not in S, $\tau \leq 1$ for all τ in H.

let $\rho = \sup H$ and $\overline{X} = (1-\rho)X_0 + \rho X_1$

then $\overline{X}$ is a limit point of S and so in S. because S is closed. since S is open. it contains some ϵ—neighborhood of $\overline{X}$, so $\rho + \frac{\epsilon}{2}$ is in H. It gives the definition of ρ.

Hence, $S = R^n$.

M.M

Q.7

Define Linear Function:-

Ans:-

A linear function is a form of function

$$L(X) = m_1x_1 + m_2x_2 + \cdots + m_nx_n$$

where $m_1, m_2, \cdots m_n$ are constant.

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Q.8

If R is degenerate, then prove by using the definition of Riemann integral that $\int_R f(X)\,dA = 0$ if f is bounded on R.

Proof:-

Let

$$R = [a_1, b_1] \times [a_2, b_2] \times \cdots \times [a_n, b_n] \text{ with } a_1 = b_1$$

then every partition P of R is of the form

$$P = [R_1, R_2, R_3, \ldots, R_K]. \text{ where } v(R_i) = 0$$

$1 \leq j \leq K$.

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Therefore,

Every Riemann sum of f over R equals zero, So

$$\int_R f(X)\,dA = 0$$

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Q.9

Determine whether the integral

$$\int_0^\infty \frac{1+\cos^2 x}{\sqrt{1+x^2}}\,dx$$

converges or diverges.

Sol:-

since

$$\frac{1+\cos^2 x}{\sqrt{1+x^2}} > \frac{1}{\sqrt{1+x^2}} = \frac{1}{x\sqrt{\frac{1}{x^2}+1}} > \frac{1}{x\sqrt{2}}$$

if $x > 1$.

M.m

since

$$\int_1^\infty \frac{dx}{x} = \infty$$

So,

$$\int_0^\infty \frac{1+\cos^2 x}{\sqrt{1+x^2}} = \infty$$

this is diverges.

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Mth631 Date:

Q.10

Calculate $\frac{\partial f(x)}{\partial \phi}$ where $f(x,y,z) = \log(1+x+y+z)$

and $\phi = (0,1,0)$.

Sol:-

We know that

$$h(t) = f(X + t\phi)$$ M.M

Given

$$f(x,y,z) = \log(1+x+y+z)$$

$$h(t) = f(x+\phi_1 t \cdot y + \phi_2 t \cdot z + \phi_3 t)$$

$$= \log(1+x+y+z+(\phi_1+\phi_2+\phi_3)t)$$

$$h'(t) = \frac{\phi_1+\phi_2+\phi_3}{1+x+y+z+(\phi_1+\phi_2+\phi_3)t}$$

at $t=0$ we get,

$$\frac{\partial f(x)}{\partial \phi} = h'(0) = \frac{\phi_1+\phi_2+\phi_3}{1+x+y+z}$$

Now put $\phi_1 = 0, \phi_2 = 1, \phi_3 = 0$

$$\frac{\partial f(x)}{\partial \phi} = \frac{0+1+0}{1+x++y+z} = \frac{1}{1+x+y+z}$$

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Mth631 Date:

Q.11

Prove that the function $f(x,y)=3x^2y^3+xy$ has eight third-order partial derivatives.

Sol:

Given $f(x,y)=3x^2y^3+xy$

1st-order partial derivatives

$f_x(x,y)=6xy^3+y$

$f_y(x,y)=9x^2y^2+x$

2nd-order partial derivatives

$f_{xx}(x,y)=6y^3$

$f_{yy}(x,y)=18x^2y$

$f_{xy}(x,y)=18xy^2+1$

$f_{yx}(x,y)=18xy^2+1$

M.M

There are eight third-order derivatives.

$f_{xxx}(x,y)=0$

$f_{yyy}(x,y)=18x^2$

$f_{xxy}(x,y)=18y^2$

$f_{xyx}(x,y)=18y^2$

$f_{yxx}(x,y)=18y^2$

$f_{yyx}(x,y)=18y^2$

$f_{yxy}(x,y)=18y^2$

$f_{xyy}(x,y)=18y^2$

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Q.12

Compute $f_{xy}(0,0)$ and $f_{yy}(0,0)$ if

$$f(x,y)=\begin{cases}\dfrac{(x^2y+xy^2)\sin(x-y)}{x^2+y^2}, & (x,y)\neq(0,0)\\ 0, & (x,y)=(0,0).\end{cases}$$

Sol:-

$$f_{xy}(0,0)=\lim_{y\to 0}\frac{f_x(0,y)-f_x(0,0)}{y}$$

$$=\lim_{y\to 0}\frac{-\sin y-0}{y}=-1$$

$$f_{yy}(0,0)=\lim_{y\to 0}\frac{f_y(0,y)-f_y(0,0)}{y}$$

$$=\lim_{y\to 0}\frac{0-0}{y}=0$$

M.M

Q.13

Maxima and Minima

Ans:-

If X_0 is a local extreme point of f if there is a $\delta>0$ such that

$$f(X)-f(X_0)$$

does not change sign in $S_\delta(X_0)\cap D_f$ then Maxima is $f(X)\leq f(X_0)$ and minima is $f(X)\geq f(X_0)$ for all X in $S_\delta(X_0)\cap D_f$

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Q.14

Show that the function $f(x,y)=x^2+2xy$ is differentiable at point (x_0,y_0).

Sol:-

$$f(x,y)-f(x_0,y_0)=x^2+2xy-x_0^2-2x_0y_0$$

$$=x^2-x_0^2+2(xy-x_0y_0)$$

M.M

$$=(x-x_0)(x+x_0)+2(xy-x_0y_0)$$

$$=(x-x_0)(x+x_0)+2(xy-x_0y+x_0y-x_0y_0)$$

$$=(x-x_0)(x+x_0)+2(xy-x_0y)+2(x_0y-x_0y_0)$$

$$=(x-x_0)(x+x_0)+2y(x-x_0)+2x_0(y-y_0)$$

$$=(x-x_0)(x+x_0+2y)+2x_0(y-y_0)$$

$$=2(x_0+y_0)(x-x_0)+2x_0(y-y_0)+(x-x_0)(x-x_0+2y-2y_0)$$

$$=m_1(x-x_0)+m_2(y-y_0)+(x-x_0)(x-x_0+2y-2y_0)$$

here $m_1=2(x_0+y_0)=f_x(x_0,y_0)$ and $m_2=2x_0=f_y(x_0,y_0)$

, by Schwarz's inequality

$$\lim_{X\to X_0}\frac{f(x,y)-f(x_0,y_0)-m_1(x-x_0)-m_2(y-y_0)}{|X-X_0|}=0$$

So, function is differentiable at (x_0,y_0).

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Mth631 Date:

Q.15

Define Riemann Sum and Reimann integral in R^n.

Ans:-

Riemann Sum in R^n

If f is a real-valued function define on a rectangle R in R^n and $P=\{R_1, R_2, \ldots, R_K\}$ is a partition of R. let X_j is an arbitrary point in R_j such that $1 \leq j \leq K$

then

M.m

$$\sigma = \sum_{j=1}^{K} f(X_j)\, V(R_j)$$

is a Reimann sum of f over P.

Riemann integral in R^n

If σ is any Riemann sum of f over a partition P of R such that $\|P\| < \delta$. here, we can say that L is the Riemann integral of f over R. and written as

$$\int_R f(X)\, dx = L$$

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Date:

Q-16

The function $f(x,y) = 1 - x^2 - 2y^2$ is continuous on R^2.

Proof:-

Give

$f(x,y) = 1 - x^2 - 2y^2$ M.M

if $f(x,y)$ is continuous on R^2 then it is continuous on all points.

So,

$f(x_0, y_0) = 1 - x_0^2 - 2y_0^2$, $x_0, y_0 \in R^2$

i.e

$f(1,-1) = 1 - (1)^2 - 2(-1)^2$

$= 1 - 1 - 2$

$= 0$

so it is continuous

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Q.17

Show that the space R^n is connected.

Proof:-

If $\mathbb{R}^n = A \cup B$ with $\bar{A} \cap B = \phi$ and $A \cap \bar{B} = \phi$, then $\bar{A} \subset A$ and $\bar{B} \subset B$. that is, A and B are both closed and therefore are both open. Since the only nonempty subset of R^n that is both open and closed is R^n itself, one of A and B is R^n and the other is empty.

M.m

Q.18

Show that $\lim_{x \to x_0} f(x)$ exists, then it is unique.

Proof:-

Let f(x) be a function.

$$\lim_{x \to x_0} f(x)$$

M.m

Then it is unique if the Domain of f(x) is,

$$D_f \subseteq \mathbb{R}^n$$

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Date:

Q.19

The function $f(x,y) = \frac{xy}{x^2+y^2}$ is defined everywhere in $\mathbb{R}^2$ except at $(0,0)$. Does $\lim_{(x,y)\to(0,0)} f(x,y)$ exist?

Sol:-

Let (x,y) approach $(0,0)$ along the line $y=x$, then function values will be

$$f(x,x) = \frac{xy}{x^2+y^2} = \frac{x\cdot x}{x^2+x^2} = \frac{x^2}{2x^2} = \frac{1}{2}$$

If (x,y) approach $(0,0)$ along the line $y=-x$, then function value will be

$$f(x,-x) = \frac{xy}{x^2+y^2} = \frac{x(-x)}{x^2+(-x)^2} = \frac{-x^2}{2x^2} = \frac{-1}{2}$$

It is clear that $f(x,x)$ and $f(x,-x)$ is not equal.

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Now

$\lim_{x\to 0} f(x,x) = \frac{1}{2}$ and $\lim_{x\to 0} f(x,-x) = -\frac{1}{2}$

$\Rightarrow \frac{1}{2} \neq -\frac{1}{2}$

Hence, $\lim_{(x,y)\to(0,0)} f(x,y)$ not exist.

It will exist if $\lim_{x\to 0} f(x,x) = \lim_{x\to 0} f(x,-x)$.

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Q.20

Show that if f is differentiable at X_0, then f is continuous at X_0.

Proof:-

We know by linear function

$$L(X) = m_1x_1 + m_2x_2 + \cdots + m_nx_n$$

by Schwarz's inequality

$$|L(X-X_0)| \leq M|X-X_0|$$

M.m

where $M = (m_1^2 + m_2^2 + \cdots + m_n^2)^{1/2}$

and

$$f(X) - f(X_0) = L(X-X_0) + E(X)(|X-X_0|)$$

imply that,

$$|f(X)-f(X_0)| \leq (M + |E(X)|)|X-X_0|.$$

when

$$\lim_{X \to X_0} E(X) = 0$$

then f is continuous at X_0.

Hence proved.

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Q.21

If S_1 has zero content and $S_2 \subset S_1$, then prove S_2 has zero content.

If $\epsilon > 0$, there are rectangles $T_1, T_2, T_3, \cdots T_r$ and $T_1', T_2', T_3', \cdots T_s'$ such that if $S_1 \subset \bigcup_{i=1}^{r} T_i$ and $S_2 \subset S_1$ then

$$S_2 \subset \bigcup_{i=1}^{r} T_i.$$

M.M

Hence S_2 has zero content.

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Q.22

If $f(s)=\sqrt{s}$ and $g(x,y)=1-x^2-2y^2$ then $D_f=[0,\infty]$, $D_g=R^2$ and $T=\{(x,y): x^2+2y^2\leq 1\}$.

Sol:-

since g is continuous on R^2

Let

$$f(g(x,y))=\sqrt{1-x^2-2y^2} \text{ and } f(s)=\begin{cases}\frac{\sin s}{s}, & s\neq 0\\ 1, & s=0\end{cases}$$

then $D_f=(-\infty,\infty)$ and

$$D_g=T=\{(x,y): x^2+2y^2\leq 1\}.$$

So the g is continuous on T.

since f is continuous on D_f, the composide function $h=fog$ defined by

$$h(x,y)=\begin{cases}\frac{\sin\sqrt{1-x^2-2y^2}}{\sqrt{1-x^2-2y^2}}, & x^2+2y^2\leq 1,\\ 1, & x^2+2y^2=1,\end{cases}$$

is continuous on

$$D_g=T=\{(x,y): x^2+2y^2\leq 1\}$$

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Q.23

rth Differential

Sol:-

If $r \geq$ and all the partial derivatives of f of order $\leq r-1$ are differentiable in a neighborhood of X_0.

Then the rth differential of f at X_0, denoted by $d^{(r)}_{X_0} f$. defined by

M.m

$$d^{(r)}_{X_0} f = \sum_{\gamma} \frac{r!}{\gamma_1! \gamma_2! \cdots \gamma_n!} \cdot \frac{\partial^r f(X_0)}{\partial x_1^{\gamma_1} \cdot \partial x_2^{\gamma_2} \cdots \partial x_n^{\gamma_n}} \cdot (dx_1)^{\gamma_1} (dx_2)$$

$$\cdot (dx_1)^{\gamma_1} (dx_2)^{\gamma_2} \cdots (dx_n)^{\gamma_n}$$

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Q.24

Show that a degenerate rectangle has zero content.

Proof:-

Let $R = [a_1, b_1] \times \cdots \times [a_n, b_n]$

and

$T = [a_1 - \epsilon, b_1 + \epsilon] \times \cdots \times [a_n - \epsilon, b_n + \epsilon]$

where $\epsilon > 0$.

Then $R \subset T$ M.M

since $a_i = b_i$ and $V(T) \leq 2\epsilon L^{n-1}$

where

$L = \max\{(b_j - a_j) \mid 1 \leq j \leq n\}$.

Q.25

If $h(u,v) = f(u^2+v^2)$, then show that $vh_u - uh_v = 0$.

$h_u = 2uf'$, $h_v = 2vf'$

Now M.M

$vh_u = 2vuf'$, $uh_v = 2uvf'$

So,

$vh_u - uh_v = 2vuf' - 2uvf'$

$= 2f'(vu - uv)$

$= 2f'(0)$

$= 0$

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